INCH-POUND

MIL-S-70843 (AR)
11 March 1993

MILITARY SPECIFICATION

SOLENOID (FOR MACHINE GUN, 7.62MM: M240)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the assembly, testing, inspection and packaging for the Solenoid (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-W-13855	- Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-W-63150	- Weapons and Support Materiel, Standard Quality Assurance Provisions for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 1005

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods for Aerospace and Ground Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12001058	-	Solenoid

PACKAGING AND MARKING

P12001058	-	Solenoid

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.2.1 and 6.2.1).

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3.2 Materials and construction. The solenoid and parts shall conform to the materials and construction requirements specified herein, on drawing 12001058 and drawings applicable thereto, and shall be in accordance with the applicable materials and construction processes of MIL-W-13855.

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, as defined in requirement 16 or MIL-STD-454; which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.3 Design. The solenoids and parts shall conform to the design specified herein, on drawing 12001058 and drawings thereto, and shall be in accordance with the applicable design provisions of MIL-W-13855.

3.3.1 Interchangeability. To provide for interchangeability, solenoids fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline on drawing 12001058. Electrical connectors, shaft and mounting dimensions and location shall also conform to the detail requirements on drawing 12001058.

3.3.2 Plunger. The plunger shall move through its full range of travel without binding.

3.4 Performance requirements.

3.4.1 Operation. Each solenoid shall operate over a voltage range of 15 to 30 volts direct current.

3.4.2 Coil resistance. The coil resistance shall be 2.30 ± 0.14 ohms at 72° ± 5°F.

3.4.3 Stroke. The solenoid stroke shall be 0.300 + 0.025 inch.

3.4.4 Duty cycle. The solenoid must be capable of operation for a period of 15 minutes consisting of 30 seconds on and 45 seconds off cycles.

3.4.5 Endurance. The solenoid shall be capable of being cycled 2 seconds on and 13 seconds off for a total of 40,000 cycles with an applied voltage of 24.0 ± 2.0 Volts DC.

3.4.6 Temperature ranges. The solenoid must be capable of operation over a temperature range of -65° ± 5°F to +160° ± 5°F.

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3.4.7 Storage temperature. The solenoid must be capable of withstanding storage temperatures of -65°F to +160°F.

3.4.8 Temperature rise. The solenoid shall withstand damage when energized continuously against a static load of not less than 30 pounds for 3.5 minutes at a temperature of +160 with an applied voltage of 30 Volts DC.

3.4.9 Waterproofness. The solenoid shall show no evidence of leakage when immersed in water to a depth of 36 inches.

3.4.10 Salt spray. The solenoid shall be capable of operation after exposure to a 5% sodium chloride atomized spray for 96 hours minimum.

3.4.11 Random vibration. The solenoid shall be capable of withstanding the random vibration level in the chart below for a period of 80 minutes (including up to 4 resonance dwells for 13 1/3 minutes each) in each of the three mutually perpendicular axes:

Acceleration Level	Frequency
0.5 Double Amplitude	5-9.9 Hz
2.5 G	10-180 Hz
2.5 G - 1.5 G	180-500 Hz

3.4.12 Shock. The solenoid shall be capable of withstanding exposure to 40 G peak, 18 ms half sine waveform, 1200 G peak, 0.7 ms half sine waveform, and 700 G peak, 0.71 ms sawtooth waveform with a velocity change of 10.2 ft/sec in both directions along each of the three mutually perpendicular axes.

3.4.13 Sand and dust. The solenoid shall be operable after 12 hours exposure to dust at velocities up to 2300 ± 500 feet per minute.

3.4.14 Fungus. The solenoid materials and assembly shall be inherently fungus resistant or treated to resist fungus growth except when used within potted or otherwise moisture sealed areas.

3.4.15 Humidity. The solenoid shall be capable of operation up to 100% relative humidity.

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3.4.16 Dielectric strength. There shall be no puncture of solid insulation, no continuing sparkover, no breakdown of the solenoid when tested.

3.4.17 Static force. The solenoid shall meet the static force requirement of drawing 12001058.

3.4.18 Return force. The return force when measured with the plunger fully retracted shall be 7.5 ± 0.5 pounds. With the plunger extended to approximately 5% short of the maximum (free) extension, the force measured at this point shall be 5.0 ± 0.5 pounds.

3.4.19 Immersion. The solenoid shall show no evidence of leakage when immersed in water to a depth of 36 inches.

3.5 Marking. Each solenoid shall be clearly marked in accordance with contract, applicable drawings and MIL-W-13855.

3.6 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-63150. In addition, the solenoid shall be free from grease, dust, rust, corrosive products, and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable

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practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-W-63150 apply and form a part of this specification. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.1.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Quality conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.2.1.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Solenoid	12001058	10

4.2.2 Tests and inspections to be performed. As required by the contract, the first article sample shall be subjected to all of the examinations and tests specified in Table I. As determined by the Government, the first article sample may be subjected to any or all requirements of the applicable drawings.

4.2.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection.

Sequence and inspections applied to each sample.

Test Sequence	Test Sample No. 1	2	3	4	5	6	7	8	9	10	Requirement Paragraph	Test Paragraph
Material and construction	X	X	X	X	X	X	X	X	X	X	3.2	4.4.1
Examination of defects	X	X	X	X	X	X	X	X	X	X	3.3	4.3.3
Fungus	X	X	X	X	X	X	X	X	X	X	3.4.14	4.4.13
Dielectric strength	X	X	X	X	X	X	X	X	X	X	3.4.16	4.4.16
Operation	X	X	X	X	X	X	X	X	X	X	3.4.1	4.4.2.1
Coil resistance	X	X	X	X	X	X	X	X	X	X	3.4.2	4.4.3
Static force	X	X	X	X	X	X	X	X	X	X	3.4.17	4.4.2.2
Temperature rise	X	X	X	X	X	X	X	X	X	X	3.4.8	4.4.7
Duty cycle	X	X	X	X	X	X	X	X	X	X	3.4.4	4.4.4
Return force	X	X	X	X	X	X	X	X	X	X	3.4.18	4.4.14
Vibration	X	X									3.4.11	4.4.10
Humidity	X	X									3.4.15	4.4.15
Shock	X	X									3.4.12	4.4.11
Sand and dust			X	X							3.4.13	4.4.12
Endurance			X	X							3.4.5	4.4.5
Immersion					X	X					3.4.19	4.4.8
Salt spray					X	X					3.4.10	4.4.9
Return force					X	X					3.4.18	4.4.14
Temperature range	X	X									3.4.6	4.4.6

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4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. The formation, size, and presentation of inspection lots shall be in accordance with MIL-W-13855. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions and contractor's/delivery schedule, but shall not exceed one month's production.

4.3.2 Examination for defects. Sample solenoids from each inspection lot shall be subjected to the examinations for defects listed in 4.3.3. Solenoids which fail to meet the specified requirements shall be rejected and removed from the lot. A random sample of solenoids shall be selected from the inspection lot in accordance with Special Inspection Levels S-4 (for major defects) and S-2 (for minor defects) of MIL-STD-105 for the examinations for defects listed in 4.3.3. Failure to meet the accept/reject criteria of 4.3.3 shall be cause for rejection of the lot.

4.3.2.1 Testing. Testing shall be as specified in 4.3.3 in the sequence listed. If any solenoid fails to meet the applicable requirements, an investigation shall be made to determine the cause of failure. Unless otherwise specified, solenoids shall be tested at ambient temperature of 72° ± 5°F.

4.3.2.2 Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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4.3.2.3 Rejected lot. After cause of failure is found, the represented lot shall be rejected subject to reconditioning and further testing as a reconditioned lot. Prior to submission as a reconditioned lot, the contractor shall screen and correct all affected solenoids of the lot. (Contractor correction shall also be effected on all solenoids not yet accepted.) A sample per 4.3.1.1 of solenoids shall be taken from the reconditioned lot and subjected to the applicable inspections. Failure of the solenoid to meet the requirements shall cause rejection of the lot.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.3.3	Solenoid		12001058
			NEXT HIGHER ASSEMBLY: N/A

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	EXAMINATIONS FOR DEFECTS	[1] Acc/Rej		
CRITICAL	None defined.			
MAJOR				
101	All parts present and assembled correctly	0 - 1	3.3	Visual
102	Placement of receptacle	0 - 1	3.3	Visual
103	Plunger travel and direction of travel	0 - 1	3.3	SMTE
104	.005 true position of .250-28UNF-2B holes (3)	0 - 1	3.3	SMTE
105	Maximum rotation of plunger, 10°	0 - 1	3.3	Visual/SMTE
106	.250-28UNF-2B thread (3) x .41 deep	0 - 1	3.3	SMTE
107	.250-28UNF-2B plunger thread x .86 deep	0 - 1	3.3	SMTE
	INSPECTIONS	[1] Acc/Rej		
MINOR				
201	4.62 maximum length of solenoid	0 - 1	3.3	SMTE
202	1.09 distance	0 - 1	3.3	SMTE
203	2.18 square max of solenoid (width by breadth)	0 - 1	3.3	SMTE
204	125 surface finish of solenoid	0 - 1	3.3	Visual
205	Protective finish	0 - 1	3.6	Visual
206	Workmanship (see MIL-W-63150)			

NOTES:

[1] See 4.3.1.1

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 2 OF 2	DRAWING NUMBER
4.3.3	Solenoid			12001058
				NEXT HIGHER ASSEMBLY
				N/A
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	PERFORMANCE TESTS			
	No. of Sample Units			
	Operation	100%	3.4.1	4.4.2.1
	Coil resistance	100%	3.4.2	4.4.3
	Static force	100%	3.4.17	4.4.2.2
	Temperature rise	100%	3.4.8	4.4.7
	Duty cycle	100%	3.4.4	4.4.4
	Return force 3		3.4.18	4.4.14
	Temperature range 3		3.4.6	4.4.6
	Endurance 3		3.4.5	4.4.5
	Vibration 3		3.4.11	4.4.10
	Shock 3		3.4.12	4.4.11
	Humidity 3		3.4.15	4.4.15

NOTES:

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4.3.4 Packaging. Examination of packaging of solenoids shall be performed in accordance with the classification of defects and acceptance criteria specified in 4.3.4.1. Sample size shall be in accordance with MIL-STD-105, using inspection level I. The following provision shall apply:

a. Examination for packaging defects specified in 4.3.4.1 shall apply to each item of the applicable sample of solenoids interior packages or exterior containers, as applicable.

4.3.4.1 Classification of defects for packaging. (Unless otherwise specified in each listed defect, the packaging requirements are specified in Section 5.)

Categories	Defect	Inspection Method
Critical	None defined.	
Major		
101.	Marking	Visual
102.	Level of packaging or packing (see procurement documents)	Visual
103.	Proper cleaning and drying	Visual
104.	Quantity of desiccant	Visual
105.	Wrapping and cushioning	Visual
106.	Closure of interior packages	Visual
107.	Proper containers	Visual
108.	Proper closure and strapping of shipping containers	Visual

4.3.5 Packaging testing.

4.3.5.1 Determination of cleanliness. The contractor shall test items from each inspection lot for cleanliness using the test methods and sample sizes specified in MIL-P-116.

4.3.5.2 Vacuum chamber. The contractor shall test level A unit packages from each inspection lot for leakage using the test methods and sample sizes specified in MIL-P-116.

4.4 Methods of inspection.

4.4.1 Materials and construction. A certificate of conformance (COC) shall be provided in accordance with MIL-W-63150 (see 6.2.1).

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4.4.2 Operation and static force test.

4.4.2.1 Operation test. Solenoids shall be tested for the operation and static force using an appropriate nonfiring test fixture (see 6.3). The solenoid shall be mounted on the test fixture and shall be operated against the specified load with an applied voltage of 15.0 + 0.5 and 30.0 + 0.0 Volts DC. The solenoid shall be cycled, 2.0 seconds on and 13.0 seconds off, for 5 minutes at each voltage. The test fixture shall be examined during the test to determine that the solenoid is producing the required stroke as specified in 3.4.3 and return force specified in 3.4.1.8.

4.4.2.2 Static force test. To test for static force the solenoid shall be operated against the specified load three times with an applied voltage of 15.0 + 0.5 Volts DC. The solenoid shall be cycled, 2.0 seconds on and 13.0 seconds off, for 5 minutes at this voltage. The test fixture shall be examined during the test to determined that the solenoid is producing the required stroke as specified in 3.4.3 and static force as specified in 3.4.1.7.

4.4.3 Coil resistance test. Solenoids shall be tested for the coil resistance. One lead of the resistance measuring device shall be connected to the pin in the solenoid receptacle and the other lead connected to a grounded portion of the solenoid.

4.4.4 Duty cycle. Solenoids shall be tested for the duty cycle requirement using the same test fixture as in 4.4.2.1.

4.4.5 Endurance test. Solenoids shall be tested for the endurance requirements using an appropriate test fixture and operated through 40,000 cycles. Each cycle consists of 2.0 seconds on and 13.0 seconds off against the specified load with an applied voltage of 24.0 ± 2.0 Volts DC. The test fixture shall be visually examined during the test to determine that the solenoid is producing the required stroke as specified on drawing 12001058.

4.4.6 Temperature range test. Solenoids shall be tested for the temperature range requirement using an appropriate test fixture (see 6.3). The solenoid shall be mounted on the test fixture, and the solenoid and the test fixture shall be placed in a temperature test cabinet with the electrical circuits connected as in intended operation.

4.4.6.1 Low temperature. The solenoid shall be tested in accordance with MIL-STD-810, Method 502, Procedure I. Maintain the temperature at -65°F for a period of 12 hours minimum. At the conclusion of this time, stabilize the assembly at -65°F and subject it

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to the performance tests specified in 4.4.2 through 4.4.4. After testing, return the assembly to 72° ± 5°F and subject to the tests specified in 4.4.2 through 4.4.4.

4.4.6.2 High temperature. The solenoid shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. Stabilize the assembly at 160°F and subject it to the tests specified in 4.4.2 through 4.4.4. After testing, return the assembly to 72° ± 5°F and subject it to the tests specified in 4.4.2 through 4.4.4.

4.4.7 Temperature rise test. Solenoids shall be tested for the temperature rise requirement by being energized continuously for 3.5 minutes ± 5 seconds with an applied voltage of 30.0 ± 0.0 Volts DC against a static load of 30 pounds and at an ambient temperature of 160° ± 5°F. At the end of 3.5 minutes, the solenoid shall be allowed to cool to 72° ± 5°F and shall be subjected to the tests specified in 4.4.2 through 4.4.4.

4.4.8 Immersion test. Solenoids shall be tested for the resistance to immersion requirements in accordance with the Method 104, Condition B, of MIL-STD-202. Not less than 4 hours and not more than 24 hours after completion of the final immersion cycle, the solenoid shall be subjected to the tests specified in 4.4.2 through 4.4.4.

4.4.9 Salt spray. Solenoids shall be tested for the resistance to salt spray requirements in accordance with the Method 101, Condition A, of MIL-STD-202 with a 5% salt solution. After exposure to salt spray, the solenoid shall be subjected to the tests specified in 4.4.2 through 4.4.4.

4.4.10 Vibration test. Solenoids shall be tested for the resistance to vibration requirement in accordance with Method 204, Condition C, of MIL-STD-202. After vibration test, the solenoid shall be subjected to the tests specified in 4.4.2 through 4.4.4.

4.4.11 Shock test. Mount the assembly in the M240 Machine Gun Mount or equivalent (see 6.3) and subject it to the shock test specified in MIL-STD-810, Method 5.6, Procedure IV. Apply three half wave shock pulses in both directions along the three mutually perpendicular axis. Peak amplitude shall be 40 ± 4 G with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. During this test, apply input power and verify that no discontinuity occurs, and that the solenoid plunger does not self actuate. Repeat the above test for 1200 G peak, 0.7 ms, half sine and 700 G peak, 0.71 ms sawtooth. After shock test, the solenoid shall be subjected to the tests specified in 4.4.2 through 4.4.4.

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4.4.12 Sand and dust test. Solenoids shall be tested for the resistance to sand and dust in accordance with Method 110, Condition A, of MIL-STD-202 except that the sand and dust velocity shall be as specified in 3.4.13.

4.4.13 Fungus. A certificate of conformance shall be provided in accordance with MIL-W-63150 (see 6.2.1).

4.4.14 Return force. To test for plunger return spring force, the solenoid shall be mounted securely on the test fixture (see 6.3). With a force gage coupled to the solenoid plunger and operating voltage removed, the plunger shall be fully retracted (operation height) then allowed to extend sufficiently to eliminate bottoming out of the plunger. The force shall be measured at this point. Continue to allow extension of the plunger to a point that is approximately 5% short of the maximum (free) extension. The force shall again be measured at this point.

4.4.15 Humidity. Place the assembly in a humidity chamber and subject to the humidity test specified in MIL-STD-810, Method 507, Procedure II. During and after this test, verify that the assembly passes the tests specified in 4.4.2 through 4.4.4.

4.4.16 Dielectric strength. Prior to final assembly, perform the dielectric strength test in accordance with MIL-STD-202, Method 301. Apply 500 V rms at a frequency of 60 Hz for a period of 75 ± 5 seconds. Verify that there is no puncture of solid insulation, no continuing sparkover and no breakdown of dielectric material.

4.5 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.3.3. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-W-63150 and 6.3 herein.

5. PACKAGING

5.1 Packing. The solenoid shall be packed in accordance with drawing P12001058.

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with drawing P12001058.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The solenoid is intended for use with the Machine Gun, 7.62mm, Coaxial M240.

6.2 Acquisition requirements. Acquisition documents must specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification and all applicable Quality Assurance Provisions (QAPs).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Quantity required and delivery schedules.

e. Serialization requirements, if applicable.

f. Quality Conformance Inspection, if other than specified in Section 4 of specification.

g. Packaging requirements, if other than specified in Section 5 of specification.

h. Certificates of conformance for each lot or shipment of product.

i. The packages opened for examination shall be repackaged by the contractor at the contractor's expense.

j. Disposition of tested solenoids.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.6 Subject term (key word) listing.

Coil Resistance
Duty Cycle
Electronic and Electrical Components
Force, Return
Weapon, Small Caliber

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1005-A804)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE		

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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